

Full Chapter Revision Worksheet

All 8 Topics · PU-II Biology, Chapter 4

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Full Chapter Revision Worksheet

16 questions across all 8 topics · answer key on the last page

- Q1.** Who is known as the "father of genetics"?
- (a) Charles Darwin
 - (b) Gregor Johann Mendel
 - (c) Thomas Hunt Morgan
 - (d) Hugo de Vries
- Q2.** Mendel conducted his hybridisation experiments on which plant?
- (a) Maize
 - (b) Garden pea (*Pisum sativum*)
 - (c) Sunflower
 - (d) Wheat
- Q3.** In a monohybrid cross TT (tall) x tt (dwarf), the F1 generation is:
- (a) All tall (Tt)
 - (b) All dwarf (tt)
 - (c) 3 tall : 1 dwarf
 - (d) 1 tall : 1 dwarf
- Q4.** On selfing the F1 (Tt x Tt), the F2 phenotypic ratio for tall:dwarf is:
- (a) 1:1
 - (b) 1:2:1
 - (c) 3:1
 - (d) 9:3:3:1
- Q5.** A dihybrid cross involves parents differing in how many pairs of contrasting characters?
- (a) One
 - (b) Two
 - (c) Three
 - (d) Four
- Q6.** In Mendel's dihybrid cross (RRYY x rryy), the F1 generation phenotype is:
- (a) Round, Green
 - (b) Wrinkled, Yellow
 - (c) Round, Yellow
 - (d) Wrinkled, Green
- Q7.** Incomplete dominance was discovered by which scientist, working on *Mirabilis jalapa*?
- (a) Gregor Mendel
 - (b) Carl Correns
 - (c) Karl Landsteiner
 - (d) Thomas Hunt Morgan
- Q8.** In incomplete dominance, the F1 phenotype:
- (a) Resembles one parent completely
 - (b) Is an intermediate/mixture of both parents, resembling neither parent
 - (c) Always resembles the mother
 - (d) Shows both parental phenotypes simultaneously, fully expressed
- Q9.** Who is known as the "father of experimental genetics"?
- (a) Gregor Mendel
 - (b) Thomas Hunt Morgan

- (c) Walter Sutton
- (d) Alfred Sturtevant

Q10. Morgan's genetic experiments were primarily carried out on which organism?

- (a) Garden pea
- (b) Guinea pig
- (c) Fruit fly (*Drosophila melanogaster*)
- (d) Honey bee

Q11. Who discovered the genetic/chromosomal basis of sex determination while studying insects, observing structures he named "X bodies"?

- (a) Thomas Hunt Morgan
- (b) Henking
- (c) Walter Sutton
- (d) Karl Landsteiner

Q12. The "X body" observed by Henking during spermatogenesis was later identified as:

- (a) An autosome
- (b) The X chromosome
- (c) The Y chromosome
- (d) A mitochondrion

Q13. Mutation is best defined as:

- (a) A change in phenotype without any change in DNA
- (b) An alteration of DNA sequence, resulting in changes in genotype and phenotype
- (c) The blending of two parental phenotypes
- (d) The independent assortment of chromosomes

Q14. Which of the following is a physical mutagen?

- (a) Nitrous acid
- (b) UV radiation
- (c) *Helicobacter* (bacteria)
- (d) HPV (virus)

Q15. Genetic disorders are broadly classified into:

- (a) Autosomal and sex-linked disorders only
- (b) Mendelian disorders and Chromosomal disorders
- (c) Dominant and recessive disorders only
- (d) Physical and chemical disorders

Q16. Mendelian disorders are caused by:

- (a) Changes in chromosome number
- (b) A mutation in a single gene, following Mendelian inheritance principles
- (c) Environmental factors only
- (d) Multiple genes acting cumulatively

Answer Key & Explanations

- Q1.** (b) Gregor Johann Mendel — Gregor Johann Mendel is called the father of genetics for his pioneering hybridisation experiments on garden pea using statistical analysis.
- Q2.** (b) Garden pea (*Pisum sativum*) — Mendel chose the garden pea, *Pisum sativum*, because it was easy to cultivate and had several contrasting characters.
- Q3.** (a) All tall (Tt) — Crossing homozygous tall (TT) with homozygous dwarf (tt) gives an F₁ that is entirely tall and heterozygous (Tt), since tall is dominant.
- Q4.** (c) 3:1 — Selfing heterozygous F₁ (Tt x Tt) gives an F₂ phenotypic ratio of 3 tall : 1 dwarf, as the recessive dwarf trait reappears.
- Q5.** (b) Two — A dihybrid cross is between two homozygous individuals differing in two pairs of contrasting characters, such as seed shape and seed colour.
- Q6.** (c) Round, Yellow — Since Round and Yellow are both dominant, the F₁ (RrYy) from RRYy x rryy shows the round, yellow phenotype.
- Q7.** (b) Carl Correns — Carl Correns discovered incomplete dominance in 1903 while working on the four o'clock plant, *Mirabilis jalapa*.
- Q8.** (b) Is an intermediate/mixture of both parents, resembling neither parent — In incomplete dominance, F₁ shows a blended, intermediate phenotype that resembles neither parent completely, unlike complete dominance or co-dominance.
- Q9.** (b) Thomas Hunt Morgan — Thomas Hunt Morgan, who worked extensively on *Drosophila melanogaster*, is called the father of experimental genetics and father of sex linkage.
- Q10.** (c) Fruit fly (*Drosophila melanogaster*) — Morgan used *Drosophila melanogaster* because it has a short life cycle, produces many progeny, and shows clear hereditary variations.
- Q11.** (b) Henking — Henking discovered the chromosomal basis of sex determination in insects, observing specific nuclear structures during spermatogenesis that he named X bodies.
- Q12.** (b) The X chromosome — The X body Henking observed was later shown to be a chromosome and was named the X chromosome.
- Q13.** (b) An alteration of DNA sequence, resulting in changes in genotype and phenotype — A mutation is an alteration of the DNA sequence itself, which can lead to changes in both genotype and phenotype.
- Q14.** (b) UV radiation — UV radiation, along with ionizing radiation and gamma rays, is a physical mutagen; nitrous acid is a chemical mutagen and HPV/*Helicobacter* are biological mutagens.
- Q15.** (b) Mendelian disorders and Chromosomal disorders — Genetic disorders are broadly classified into Mendelian disorders, caused by mutation in a single gene, and Chromosomal disorders, caused by changes in chromosome number or structure.
- Q16.** (b) A mutation in a single gene, following Mendelian inheritance principles — Mendelian disorders arise from mutation in a single gene and follow the classic principles of Mendelian inheritance.